PRIYANKA BANA

+91-7020767107 | priyankabana0408@gmail.com | linkedin.com/in/priyankabana | github.com/priyankabana | priyankabana.vercel.app

Education

Vishwakarma University — Pune, MH
Bachelor of Technology in Computer Engineering; CGPA: 7.77 (through Semester 6) — *2023 to 2027*

Experience

Software Development Intern, Data and AI — Jul 2025 to Present
BuySellTips Financial Research Pvt. Ltd. — *Pune, MH*

- Own the automated research-report pipeline that turns SEC filings and earnings-call transcripts into published equity research for **US (NYSE, NASDAQ, AMEX, ADR) and Australian (ASX)** companies, orchestrating parallel **AWS Lambda** stages over **Neo4j, Qdrant and PostgreSQL** with cron scheduling and emailed run reports.
- Led the **ASX market launch** end to end: found the 4 blockers that had silently stopped earlier runs (US-only selectors, half-year period labels, an unsafe graph rewrite, an ignored date parameter), added HY/FY support and a stale-data guard, and shipped QC-gated reports with human review before publishing.
- Built a 10-Q competitive-intelligence report product for **500+ US large caps** from spec to production: SQL peer selection, rule-based rejection before any LLM spend, a content-hash extraction cache that makes re-runs **about 90% cheaper**, automated QC of tables and figures, and offline tests.
- Built an LLM quality gate that checks **100% of reports** instead of samples (14 code rules plus a 57-question classifier routing to block, review or publish); on a hand-labelled golden set it raised accuracy from **81% to 90%** and **halved reviewer workload** at about 1 cent per report.
- Cut LLM cost per report **about 60%** by retrieving filing context from Qdrant under a token budget instead of sending whole filings, and cut transcript ingestion from **about 45 min to 2 min**.
- Rebuilt the earnings-calendar loader across **6 countries** (US, Australia, India, UK, Germany, Canada) processing about 96K events per run: raised the match rate from **23% to 96% across 1,200+ companies** with a 3-source fallback, removed about 27K stale revisions per run, and traced recurring overnight corruption to an undocumented **AWS Fargate** scraper.
- Root-caused recurring "no filing" rejects to two Lambdas parsing fiscal years differently for non-December fiscal-year companies, plus a retry deadlock; shipped idempotent, cross-checked graph repairs and found that **87% of stored transcripts were empty** due to a parser schema mismatch.

Projects

Shared EC2 Scheduler | *Bash, flock, AWS EC2, cron* — *Work project, BuySellTips*

- Lease system that lets 3 daily jobs (IPO, calendar, transcripts) share one **EC2** instance safely; a reaper guarantees it is never left idle more than about 30 min. Verified with **60 concurrent interleavings** in simulation, zero violations.

Serverless Bond-Price Pipeline | *AWS Lambda, SQS, EventBridge, ECR, PostgreSQL* — *Work project, BuySellTips*

- Collects daily FINRA price and yield data for US bonds with container **Lambdas** fed by **SQS** with a dead-letter queue; one transaction per batch so retries never create duplicates.

Touchstone: Self-Verifying Financial Data Pipeline | *Python, FastAPI, SQLite, OpenAI API, pytest*

- Built a verification layer for SEC EDGAR 13F data that proves, auto-corrects or holds each row, catching **5 of 5 planted faults** (ticker renames, a 1,302x currency-unit error, stale feeds, amended filings). LLM-proposed checks go live only after AST allowlisting, sandboxed replay and human approval.

GeoTrade: Supply-Chain Exposure Map | *Python, FastAPI, PostgreSQL, PostGIS, React, Docker*

- Consolidated **29,442 raw listings** from 6 countries' exchanges into **23,020 unique, geocoded companies** (0 validation errors), served by a 25-table PostGIS schema and a FastAPI service measured at 2–4 ms against a 200 ms budget.

HeyVanilla: Personal Productivity Platform | *Next.js 14, TypeScript, Supabase, Docker*

- Built a 15+ module full-stack app with row-level security on all 20 tables, session middleware and CSP/HSTS headers.

Technical Skills

Languages: Python, Java, SQL, TypeScript, JavaScript, Bash | **Core**: DSA, OOP, DBMS, OS, Computer Networks
Backend and DBs: FastAPI, Flask, Django, Spring Boot, Node.js, Next.js, PostgreSQL, MySQL, Neo4j, Qdrant, MongoDB
Cloud and Tools: AWS (Lambda, EC2, S3, SQS, EventBridge, ECR, Fargate), Docker, Linux, Git, pytest, Selenium, OpenAI and Anthropic APIs